Feeling Generous Under Uncertainty *

Brandon Zhou †

April 24, 2026

Abstract

This study examines whether uncertainty influences both prosocial behavior and individuals' perceptions of their own generosity. Participants first complete a baseline dictator game, after which they are randomly assigned to either an uncertainty or control condition before making additional allocation decisions. Following these decisions, participants report their self-perceived generosity, both in absolute terms and relative to others. Consistent with prior work, uncertainty does not significantly affect self-perceived generosity across measures, although perceived generosity varies systematically with baseline giving behavior. These results suggest that while uncertainty may influence decision-making in some contexts, individuals' evaluations of their own past actions remain anchored to their observed behavior. These findings have implications for understanding prosocial behavior in real-world settings, where outcomes are often uncertain, and for identifying the psychological mechanisms linking behavior to moral self-image.

*This paper was written as part of an undergraduate honors thesis in Behavioral Economics. Special thanks to my Thesis Advisor, Peter Schwardmann. All mistakes are my own. The most recent version of the paper and replication materials are available at: `https://github.com/brazhou04/honors-thesis/`.

†Carnegie Mellon University. Email: brandonz@andrew.cmu.edu.

1 Introduction

Understanding how people make decisions under uncertainty is critical because uncertainty encompasses nearly all consequential domains of human behavior, ranging from financial investments (Kahneman and Tversky, 1979) to medical ethics (Redelmeier and Shafir, 1995) and moral judgment (Tobia et al., 2013). Although classical decision theory treats uncertainty as a rational calculus of probabilities and utilities (Savage, 1954), behavioral research increasingly shows that people's responses to uncertainty often diverge from purely rational models. A recent study by Chen and Zhong (2024) proposes a model of decision-making under uncertainty that integrates elements of complexity and anxiety (Chen and Zhong, 2024). They found that participants behave more prosocially under uncertainty, giving more compared to their control counterparts in the Dictator Game. They argue that under such conditions, individuals may rely on a form of "quasi-magical thinking" Shafir and Tversky (1992) to evaluate the utility of perceived moral or immoral actions.

Chen and Zhong also found that participants who were asked to predict others' decisions also were similarly affected by uncertainty, suggesting that the driver of these effects lies in a general shift in beliefs or evaluative processes under uncertainty. The norm of behavior appears to shift to be more prosocial under uncertainty, consistent with Chen and Zhong's findings. This raises the question of whether such shifts in perceived norms extend beyond expectations about others' behavior to influence how individuals interpret and evaluate their own past actions.

Psychological studies have demonstrated that uncertainty can alter both the structure of choices and the processes used to evaluate them (Hirsh et al., 2012), influencing not only outcomes but also the underlying motivations, such as whether individuals rely more on deliberative reasoning or on emotional heuristics. This is consistent with the hot-cold empathy gap, which suggests that individuals in a "cold," emotionally neutral state systematically underestimate the influence of visceral or affective states on their own behavior and preferences (Loewenstein, 1996). A key distinction however, is that these findings are about

a prediction mismatch, rather than changes in belief towards past actions, as described by self-perception theory, which maintains that individuals infer internal states from observed behavior (Bem, 1972). Outcome bias shows that individuals may systematically revise their evaluations of past decisions when new outcome information becomes available, even when the decision itself is unchanged (Baron and Hershey, 1988), suggesting that retrospective judgment can be influenced by post hoc information that is diagnostic of the action.

Despite substantial evidence that uncertainty and affective states influence prospective decision-making, much less is known about how these findings relate to retrospective self-evaluation. In many settings, individuals are not only required to make decisions under uncertainty, but also to interpret and evaluate their own past behavior after uncertainty has been introduced or resolved. For example, individuals may make financial or moral decisions in one context, and later reassess those actions when faced with new uncertainty about outcomes. Whether such post hoc uncertainty can reshape self-perception has important implications for our understanding of moral self-concept, accountability, and belief updating.

This distinction raises a key question: does uncertainty merely influence behavior at the moment of choice, or can it also alter how individuals perceive and evaluate actions they have already taken? While prior work suggests that affective states can bias judgment and decision processes, it remains unclear whether these effects are strong enough to reframe recent, concrete behavior, particularly when that behavior is directly observable and salient to the individual. Theories such as self-perception suggest that individuals anchor their evaluations in their own actions, whereas research on affect and uncertainty implies that cognitive and emotional states may still shape judgment.

If post hoc uncertainty were found to alter perceptions of past behavior, this would suggest that mechanisms such as quasi-magical thinking extend beyond decision-making into retrospective self-evaluation, which would clarify how individuals construct and revise moral self-beliefs under uncertainty. This would also imply that shifts in perceived norms can retroactively shape how individuals interpret their own actions, rather than operating

only at the point of decision.

This study examines whether an uncertainty manipulation applied after a social decision influences individuals' perceived generosity, holding the actual behavior constant. By separating the timing of decision and evaluation, this design isolates whether uncertainty operates solely at the point of action or also affects retrospective interpretation. Understanding this distinction provides insight into the temporal boundaries of affect-driven judgment, and helps clarify whether mechanisms such as quasi-magical thinking extend beyond decision-making into self-perception.

2 Experimental Design

This project uses a between-subjects experimental design to examine whether uncertainty affects how people perceive their own generosity, rather than generosity alone. Participants (N = 102) were recruited through Prolific and completed the study online via Qualtrics. Participants who failed comprehension checks or had incomplete responses were excluded, yielding a final analytic sample of 77 participants.

Participants first completed a standard dictator game to establish a baseline measure of prosocial behavior. In this task, participants were endowed with 10 points and asked to allocate any integer amount between themselves and an anonymous recipient, who had no decision-making power. Participants were told that their decision could be picked, where one point was equivalent to $1 in potential bonus compensation; this system was used to maintain incentive compatibility while keeping total study costs manageable. The dictator game is a canonical paradigm in experimental economics for measuring altruistic or prosocial preferences, as it isolates unilateral giving without strategic considerations (Forsythe et al., 1994; Camerer, 2003). Collecting this measure prior to any manipulation allows the study to control for stable individual differences in baseline generosity.

After completing the dictator game, participants were randomly assigned to either an

uncertainty condition or a control condition. In both conditions, participants completed a second task involving a choice among six boxes. In the control condition, all boxes contained a fixed payoff of $2, eliminating outcome uncertainty. In the uncertainty condition, one box contained $5.75 and the remaining boxes contained $1.25, with equal probability across boxes, introducing outcome uncertainty while holding expected value constant. This design was based entirely on Chen and Zhong's 2024 study, with costs adjusted to fit budget constraints. Participants were informed that payments would be given after the survey.

Following this manipulation, participants completed a self-report questionnaire on perceived generosity. The primary outcome measure consists of three items measured on 7-point Likert scales: (i) a relative self-assessment ("Compared to others in your position, I believe I gave..."), (ii) a fairness evaluation of their allocation decision, and (iii) a belief about the average giving of others in the task. These measures were intended to elicit participants' beliefs about their generosity and the perceived norm of giving under uncertainty. Five additional questions were included to measure magical ideation, using items adapted from the Magical Ideation Scale developed by Eckblad and Chapman (1983). The three primary outcome measures were compared across treatment conditions while controlling for baseline dictator game behavior. Data analysis was conducted in Python. An analysis of variance (ANOVA) framework was used to estimate the effect of uncertainty on perceived generosity, including controls for allocation category. This approach allows the study to distinguish between two competing interpretations: whether uncertainty affects behavior itself or instead shifts how individuals interpret and evaluate their own actions. Replication materials, including all analysis code and a description of the survey design, are available at `https://github.com/brazhou04/honors-thesis/`.

Table 1: Summary Statistics

Variable	Mean	Std. Dev.	Min	Max	N
actual_gave	3.708	2.025	0.000	10.000	77
gave_share	0.371	0.203	0.000	1.000	77
compare_num	4.299	1.338	1.000	7.000	77
fair_num	5.584	1.757	1.000	7.000	77
average_guess	3.662	1.353	0.000	7.000	77
others_minus_self	-0.045	2.248	-8.000	5.000	77
magic_raw	3.743	1.476	1.000	6.600	77

3 Results

Table 1 reports summary statistics for the main variables. Participants allocated an average of 3.71 out of 10 units in the dictator game (SD = 2.03), corresponding to 37.1% of the endowment. Self-reported generosity (mean = 4.30) and fairness (mean = 5.58) were above the midpoint of the scale, while beliefs about others' giving were somewhat lower (mean = 3.66). The norm-gap measure, defined as the difference between expected giving by others and one's own allocation, is centered near zero on average (mean = -0.045) but exhibits substantial variation (SD = 2.25), indicating heterogeneity in how participants perceive their behavior relative to others.

Table 2: Treatment and Giving as Predictors of Self-Evaluation

	Relative Self-Assessment	Fairness Rating	Belief About Average Giving
	(1)	(2)	(3)
Share Given	4.657***	6.037***	1.140
	(0.555)	(1.373)	(0.880)
Treatment	-0.016	0.231	0.485
	(0.221)	(0.280)	(0.299)
Intercept	2.580***	3.232***	3.000***
	(0.286)	(0.514)	(0.385)
Observations	77	77	77
R^2	0.497	0.483	0.058
Adjusted R^2	0.483	0.469	0.033
Notes:			*p<0.1; **p<0.05; ***p<0.01

Table 2 presents regression results examining the relationship between treatment, giving behavior, and perceived generosity. Across all specifications, participants' own behavior

is a strong and statistically significant predictor of self-evaluation. A one-unit increase in the share of the endowment given is associated with a 4.66-point increase in relative self-assessment ($p < 0.01$) and a 6.04-point increase in fairness ratings ($p < 0.01$). These effects are large and precisely estimated, indicating that individuals strongly anchor their self-perceptions in their observed behavior.

In contrast, the treatment effect is small and statistically insignificant across all outcomes. The coefficient on treatment is close to zero for relative self-assessment ($\beta = -0.016$) and remains insignificant for fairness ($\beta = 0.231$) and beliefs about others ($\beta = 0.485$). These findings provide no evidence that the uncertainty manipulation alters how participants evaluate their own generosity.

The explanatory power of the models differs across outcomes. The models explain a substantial portion of variation in self-assessment and fairness ($R^2 \approx 0.50$), but much less variation in beliefs about others ($R^2 = 0.058$), suggesting that beliefs about others are noisier and less tightly linked to individual behavior.

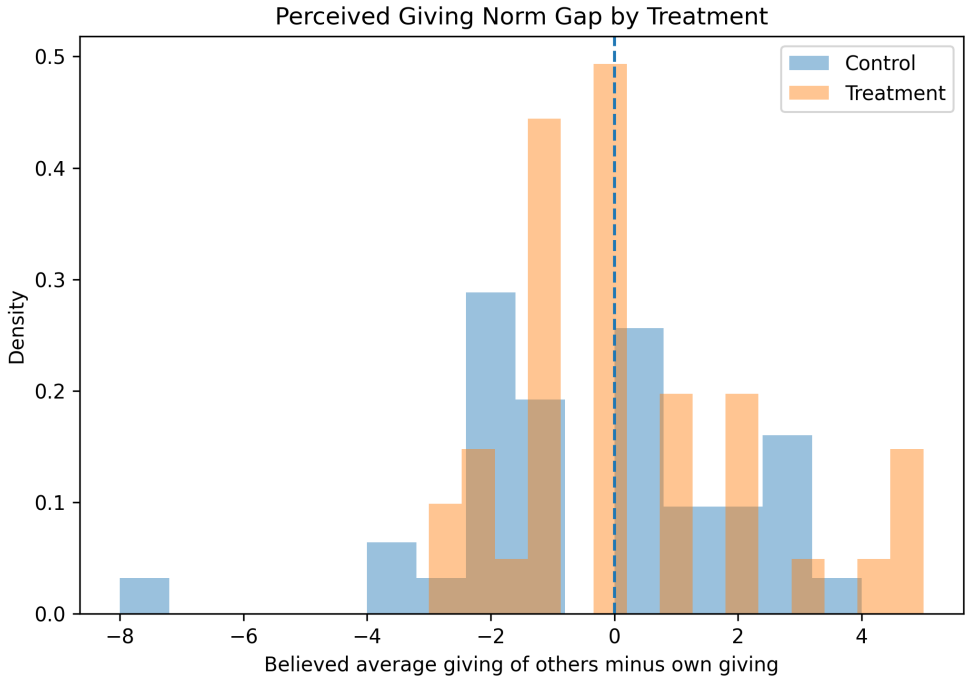

Figure 1: Distribution of the perceived norm gap (others minus self) by treatment condition.

Figure 1 illustrates the distribution of the norm-gap measure across treatment conditions. The distributions are broadly similar between the control and uncertainty conditions, consistent with the regression results indicating no significant treatment effect. However, the wide spread of the distribution highlights substantial individual heterogeneity in perceived norms.

4 Discussions

The results of this study indicate that the uncertainty manipulation did not significantly affect participants' perceived generosity across any of the three self-report measures. An additional insight from the results concerns participants' beliefs about the behavior of others. Across both the regression and ANOVA analyses, the uncertainty manipulation did not significantly affect participants' estimates of average giving, nor did it meaningfully interact with their own behavior. This suggests that the perceived social norm of generosity remains largely stable, even when individuals are exposed to uncertainty.

Importantly, the past action seems to outweigh the potential change in norms induced by the uncertainty condition. This finding aligns with self-perception theory (Bem, 1972), which suggests that people use their own behavior as primary evidence when forming self-judgments. When behavior is recent and unambiguous, it may dominate over shifts in context or affect, limiting retrospective reinterpretation.

These findings have important implications for interpreting the null treatment effect. One potential channel through which uncertainty could have influenced self-perception is by altering beliefs about the reference group, effectively shifting the benchmark against which individuals evaluate their own actions. However, the absence of any detectable change in average_guess indicates that this channel is not operative in the present setting.

Taken together, these findings suggest that both components of self-evaluation, one's own behavior and the perceived norm, are resistant to short-term contextual manipulations

introduced after a decision has been made. While individuals strongly anchor their self-perceptions in their own actions, the reference point provided by beliefs about others remains fixed.

A plausible explanation for the results is related to the temporal relationship between behavior and self-evaluation. Participants completed the dictator game before the uncertainty manipulation, meaning that by the time they were asked to reflect on their generosity, their allocation decision was already fixed. It is possible that people's evaluations of recent past actions are resistant to reinterpretation, particularly when the decision was made only moments before. Unlike prospective decisions, where uncertainty about future outcomes may activate quasi-magical thinking or moral self-licensing Shafir and Tversky (1992); Chen and Zhong (2024), retrospective evaluations of behavior may be more anchored to the past action itself. They know what they did, thus the salience of the prior act may dominate over any uncertainty induced afterward.

It may also be the case that there was not enough temporal distance within the experimental design, given the short interval between the dictator game and the self-report measure, since participants completed the two tasks back-to-back. If affect-driven reinterpretation of past behavior requires sufficient temporal distance, it may simply not have been long enough for the manipulation to alter recall.

A second, more methodological-based interpretation is that the manipulation itself may not have successfully induced uncertainty. If participants in the uncertainty condition did not experience meaningfully elevated uncertainty relative to controls, any null effect on perceived generosity would be expected. Without a direct manipulation check measuring subjective uncertainty, it is difficult to rule out this possibility.

Finally, it is worth noting that the null result may simply reflect limited statistical power. With the sample available, the study may have been underpowered to detect a small but real effect of uncertainty on perceived generosity. Replication with a larger sample and a pre-registered manipulation check would help clarify whether the absence of an effect is

theoretically meaningful or a function of design constraints.

Taken together, these findings suggest that uncertainty may primarily influence prospective prosocial behavior rather than retrospective self-evaluation. The mechanisms proposed by Chen and Zhong (2024), including complexity, anxiety, and quasi-magical thinking, are likely to be the most active at the point of decision, not after. Future research might directly compare uncertainty manipulations administered before versus after a prosocial decision to test whether timing moderates the effect on self-perception, as well as a validation that the treatment induced uncertainty.

5 Conclusion

This paper examined whether uncertainty affects how individuals evaluate their own generosity after making a prosocial decision. Using a between-subjects experimental design, participants first completed a dictator game and were then exposed to either an uncertainty or control condition before reporting their perceptions of generosity, fairness, and beliefs about others.

The results provide no evidence that uncertainty alters retrospective evaluation. Across all specifications, the uncertainty manipulation had no statistically significant effect on perceived generosity, fairness, or beliefs about the behavior of others. Higher levels of giving are associated with higher self-assessed generosity and fairness.

In addition, beliefs about others remained largely unchanged across conditions, suggesting that perceived social norms are stable and not easily shifted by short-term contextual manipulations.

These results contribute to a growing literature on the role of uncertainty in decision-making by highlighting an important temporal distinction. While uncertainty may influence behavior at the point of choice, its effects appear limited once a decision has been made. Mechanisms such as complexity, anxiety, and quasi-magical thinking, as proposed by Chen

and Zhong (2024), are therefore likely to operate primarily in prospective contexts.

Several limitations should be noted. The study may have been underpowered to detect small effects, and the absence of a direct manipulation check limits the ability to confirm whether the uncertainty treatment successfully altered participants' subjective experience. Additionally, the short temporal gap between decision and evaluation may have reduced the likelihood of post hoc reinterpretation.

Future research could build on these findings by varying the timing of uncertainty manipulations, introducing delays between behavior and evaluation, and incorporating direct measures of perceived uncertainty. Such extensions would help clarify the conditions under which uncertainty influences not only what people do, but how they interpret what they have already done.

Overall, the findings suggest that uncertainty does not meaningfully reshape how individuals perceive their own generosity after acting. Instead, self-evaluation appears to rely on stable behavioral anchors, reinforcing the idea that actions themselves serve as the primary basis for self-judgment.

References

Baron, J. and Hershey, J. C. (1988). Outcome bias in decision evaluation. *Journal of Personality and Social Psychology*, 54(4):569–579.

Bem, D. J. (1972). Self-perception theory. *Advances in Experimental Social Psychology*, 6:1–62.

Camerer, C. (2003). *Behavioral Game Theory: Experiments in Strategic Interaction*. Princeton University Press.

Chen, Y. and Zhong, S. (2024). The effect of uncertainty on prosocial behavior. *Econometrica*, 92(2).

Forsythe, R., Horowitz, J. L., Savin, N. E., and Sefton, M. (1994). Fairness in simple bargaining experiments. *Games and Economic Behavior*, 6(3):347–369.

Hirsh, J. B., Mar, R. A., and Peterson, J. B. (2012). Psychological entropy: A framework for understanding uncertainty-related anxiety. *Psychological Review*, 119(2):304–320.

Kahneman, D. and Tversky, A. (1979). Prospect theory: An analysis of decision under risk. *Econometrica*, 47(2):263–291.

Loewenstein, G. (1996). Out of control: Visceral influences on behavior. *Organizational Behavior and Human Decision Processes*, 65(3):272–292.

Redelmeier, D. A. and Shafir, E. (1995). Medical decision making in situations that offer multiple alternatives. *JAMA*, 273(4):302–305.

Savage, L. J. (1954). *The Foundations of Statistics*. Wiley.

Shafir, E. and Tversky, A. (1992). Thinking through uncertainty: Nonconsequential reasoning and choice. *Cognitive Psychology*, 24(4):449–474.

Tobia, K., Buckwalter, W., and Stich, S. (2013). Moral intuitions: Are philosophers experts? *Philosophical Psychology*, 26(5):629–638.